

EAGLE RIVER, WI, USA

WMO: 726504

Lat: 45.932N

Lon: 89.269W

Elev: 501

StdP: 95.46

Time Zone: -6.00 (NAC)

Period: 97-19

WBAN: 54838

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-27.2	-23.6	-31.0	0.2	-26.3	-27.6	0.3	-22.6	9.1	-7.8	8.3	-8.6	1.0	260	0.580

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.8	29.0	20.5	27.6	19.6	26.3	18.7	22.8	26.3	21.6	25.1	20.6	24.0	4.1	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.1	17.8	24.7	20.9	16.5	23.5	19.1	14.8	22.1	69.9	26.4	65.7	25.4	61.8	24.4	27.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.6	7.6	6.8	DB	-31.4	31.8	1.9	1.6	-32.7	33.0	-33.8	34.0	-34.9	34.9	-36.2	36.1	(4)
(5)			WB	-31.5	24.3	1.9	1.6	-32.9	25.5	-34.0	26.4	-35.0	27.3	-36.4	28.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	4.8	-10.3	-9.2	-3.2	4.2	11.2	16.3	18.7	17.3	13.5	6.4	-0.6	-7.5	(6)	
	DBStd	11.61	6.89	6.84	6.95	5.38	5.00	3.97	3.31	3.40	4.66	4.87	5.39	6.44	(7)	
	HDD10.0	2824	630	537	413	184	45	3	0	0	18	135	319	541	(8)	
	HDD18.3	5049	888	770	668	423	226	84	37	61	156	369	568	799	(9)	
	CDD10.0	936	0	0	3	11	83	192	269	227	124	25	1	0	(10)	
	CDD18.3	118	0	0	0	1	6	23	48	28	12	1	0	0	(11)	
	CDH23.3	1226	0	0	1	9	96	251	494	261	108	6	0	0	(12)	
	CDH26.7	234	0	0	0	1	19	46	107	45	17	0	0	0	(13)	
(14) Wind	WSAvg	2.7	2.8	2.9	2.8	3.1	3.0	2.5	2.3	2.1	2.4	2.7	2.8	2.8	(14)	
(15) Precipitation	PrecAvg	801	30	24	45	66	91	98	108	82	94	84	47	34	(15)	
	PrecMax	1035	74	50	108	146	189	200	206	158	209	160	127	63	(16)	
	PrecMin	618	7	1	10	19	31	51	36	38	21	29	10	8	(17)	
	PrecStd	104	18	13	25	34	37	36	44	33	44	39	26	15	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.8	9.9	19.9	24.7	29.0	30.3	31.3	30.2	28.8	23.9	16.4	7.8	(19)	
		MCWB	3.1	5.4	13.6	14.8	18.4	21.0	22.7	21.0	21.1	16.7	11.2	5.4	(20)	
	2%	DB	2.6	6.0	13.8	20.2	26.2	27.8	29.1	27.7	26.3	21.1	12.7	4.3	(21)	
		MCWB	1.2	3.2	8.8	11.2	16.2	19.5	20.9	20.1	18.9	15.8	9.3	2.3	(22)	
	5%	DB	0.9	2.8	10.2	17.3	23.0	26.3	27.7	26.2	24.0	17.7	10.0	2.3	(23)	
		MCWB	0.3	1.0	6.7	9.1	14.3	18.3	20.0	19.2	17.9	13.7	6.8	1.1	(24)	
	10%	DB	-1.0	0.3	7.0	14.1	21.0	24.2	26.3	24.4	22.2	15.1	7.5	0.9	(25)	
		MCWB	-1.7	-0.9	3.8	7.6	13.6	17.1	19.0	18.2	17.1	11.2	5.2	0.2	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.9	6.5	14.6	16.2	21.1	23.4	24.9	23.6	22.2	20.8	12.5	6.4	(27)	
		MCDB	4.5	7.8	18.0	21.1	25.8	27.8	29.2	26.6	25.7	22.0	14.3	7.4	(28)	
	2%	WB	1.3	3.3	10.9	12.8	18.7	21.6	23.2	22.1	20.8	16.9	10.2	2.9	(29)	
		MCDB	2.5	5.4	12.2	18.0	22.7	25.7	26.7	25.1	24.0	18.7	12.5	3.9	(30)	
	5%	WB	0.1	1.3	6.8	10.3	16.9	20.3	21.9	20.9	19.5	14.1	7.3	1.2	(31)	
		MCDB	0.9	2.9	9.6	15.8	20.5	24.0	25.5	24.0	22.5	17.3	9.0	2.0	(32)	
	10%	WB	-1.7	-0.9	3.7	8.2	15.0	18.9	20.6	19.8	18.1	12.0	5.3	0.1	(33)	
		MCDB	-1.2	0.1	6.4	13.2	19.2	22.3	24.1	23.3	20.6	14.6	7.3	0.8	(34)	
(35) Mean Daily Temperature Range	MDBR		9.0	10.9	11.7	12.1	13.2	12.4	12.8	12.4	12.1	10.1	7.9	7.9	(35)	
	5% DB	MCDBR	9.8	11.8	13.9	18.2	16.3	14.3	14.0	13.7	14.3	14.3	12.6	9.3	(36)	
		MCWBR	8.4	8.9	8.8	9.9	7.6	7.0	6.8	6.9	7.5	8.9	8.8	7.6	(37)	
	5% WB	MCDBR	9.2	10.6	12.4	15.3	12.5	12.1	11.9	11.3	11.2	12.1	11.0	7.8	(38)	
		MCWBR	8.1	8.0	8.6	9.7	7.2	7.0	6.8	6.6	7.0	8.8	8.5	6.8	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.239	0.266	0.309	0.332	0.363	0.363	0.362	0.360	0.342	0.306	0.279	0.249	(40)	
	taud		2.396	2.311	2.344	2.430	2.388	2.406	2.431	2.436	2.488	2.578	2.546	2.442	(41)	
	Ebn at Noon		892	920	922	928	904	902	898	887	878	871	836	839	(42)	
	Edh at Noon		69	93	106	107	116	114	110	106	92	73	62	60	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.42	2.46	3.72	4.56	5.38	5.81	6.01	5.07	3.90	2.39	1.38	1.01	(44)	
	RadStd		0.14	0.27	0.35	0.46	0.45	0.46	0.30	0.32	0.24	0.26	0.15	0.11	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	-0.99	N/A	N/A	N/A	N/A	N/A
(47)	Regional (36 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page